

From Diagnosis to Action: Advancing Sickle Cell Care in Sierra Leone

SICKLED
Advocacy for Sickle Cell Education and Diagnosis

Our Team



Amelie Fry '27
IDEAS



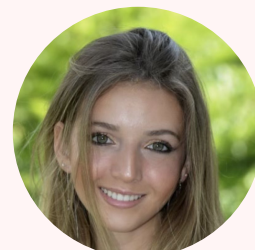
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Bioengineering



Nancy Carcamo Paiz '27
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IDEAS



Jake Feuerstein '26
IDEAS



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Lauryn Jones '25
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Biology

Faculty Advisors: Dr. Xuanhong Cheng and Professor Khanjan Mehta

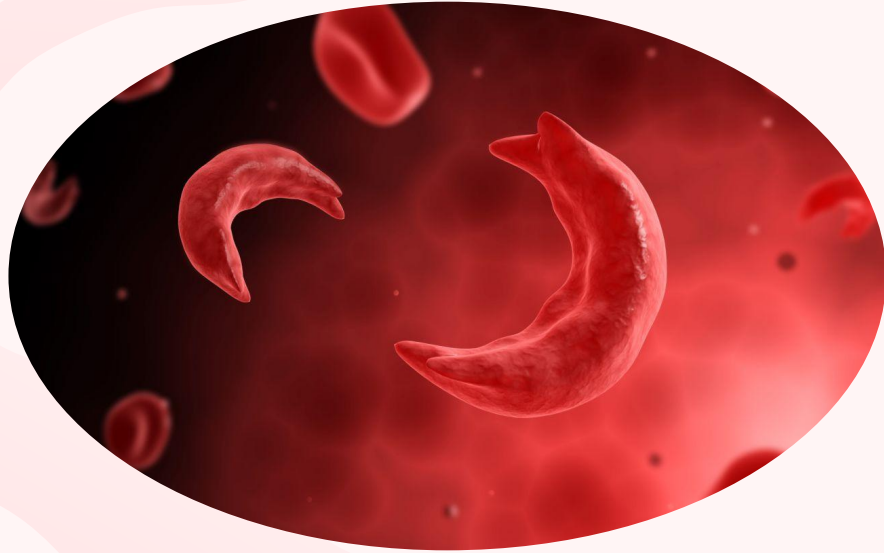
Our Partners

United States



Sierra Leone





The Problem

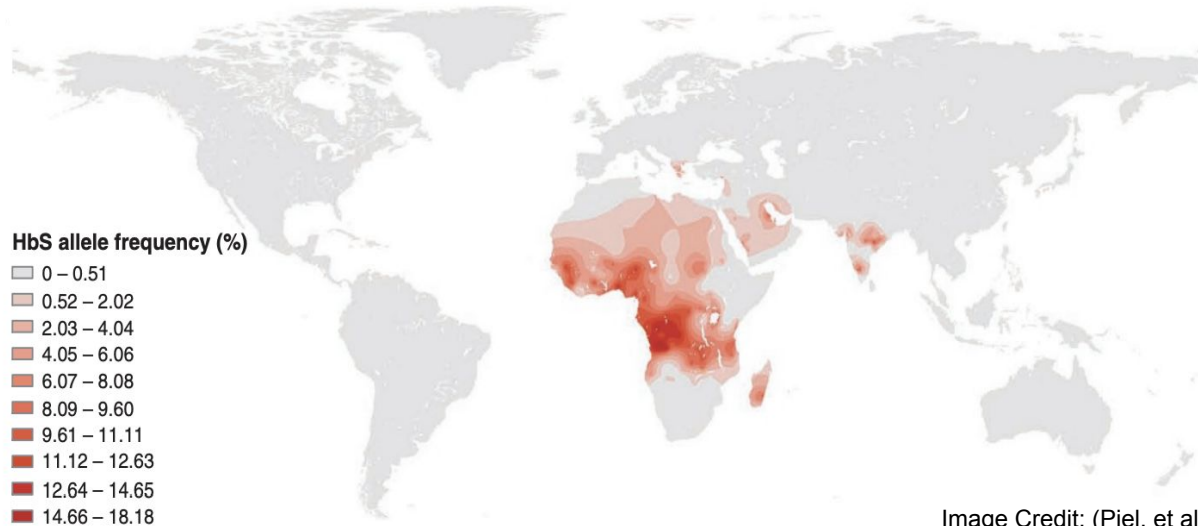
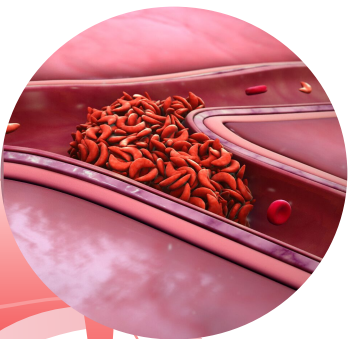
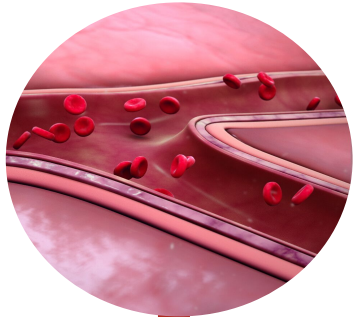


Image Credit: (Piel, et al., 2010)

**300,000 children
born with SCD Each
Year** (*Tshilolo, 2019*)

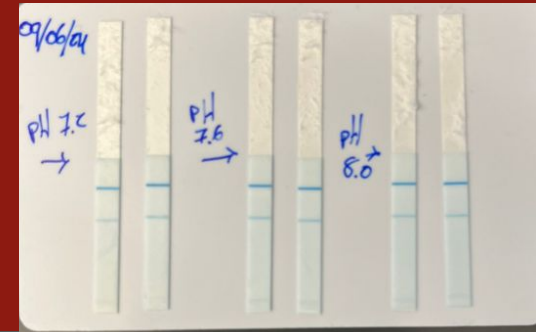
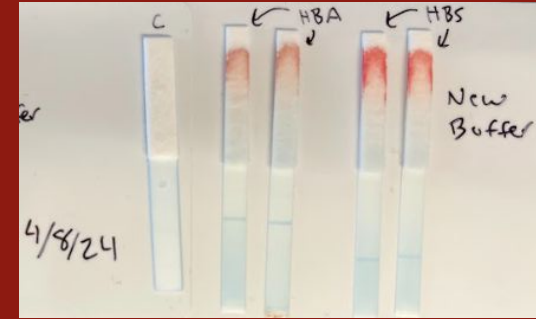
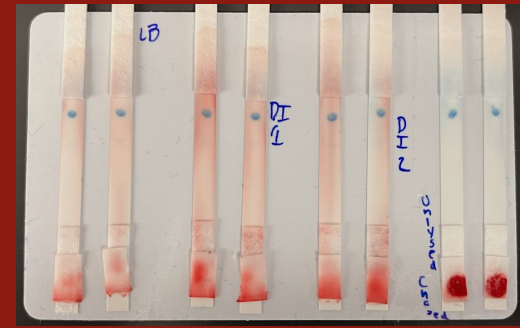
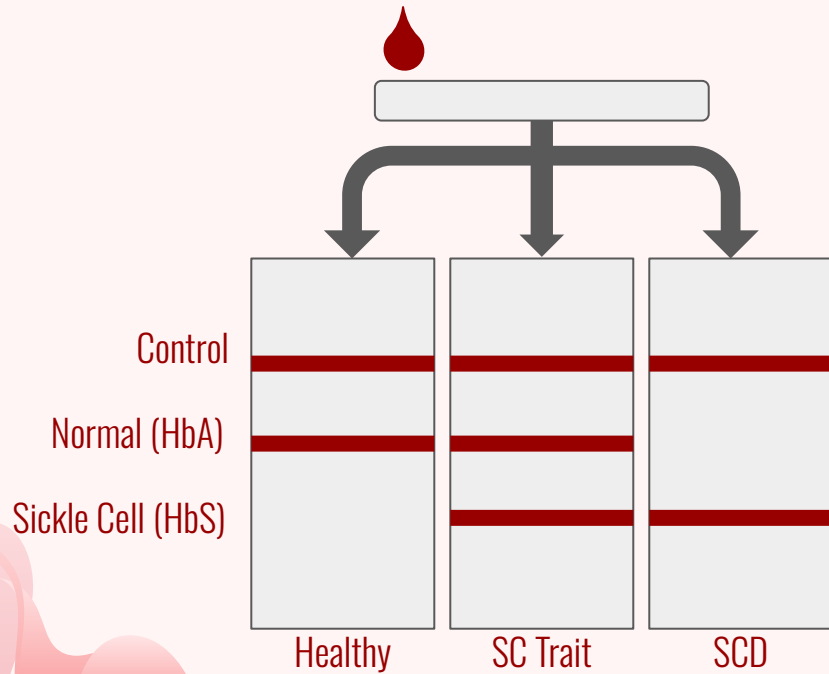
The majority of these births occur in Sub-Saharan Africa or where Malaria Incidence is also high (Piel et. al., 2010)

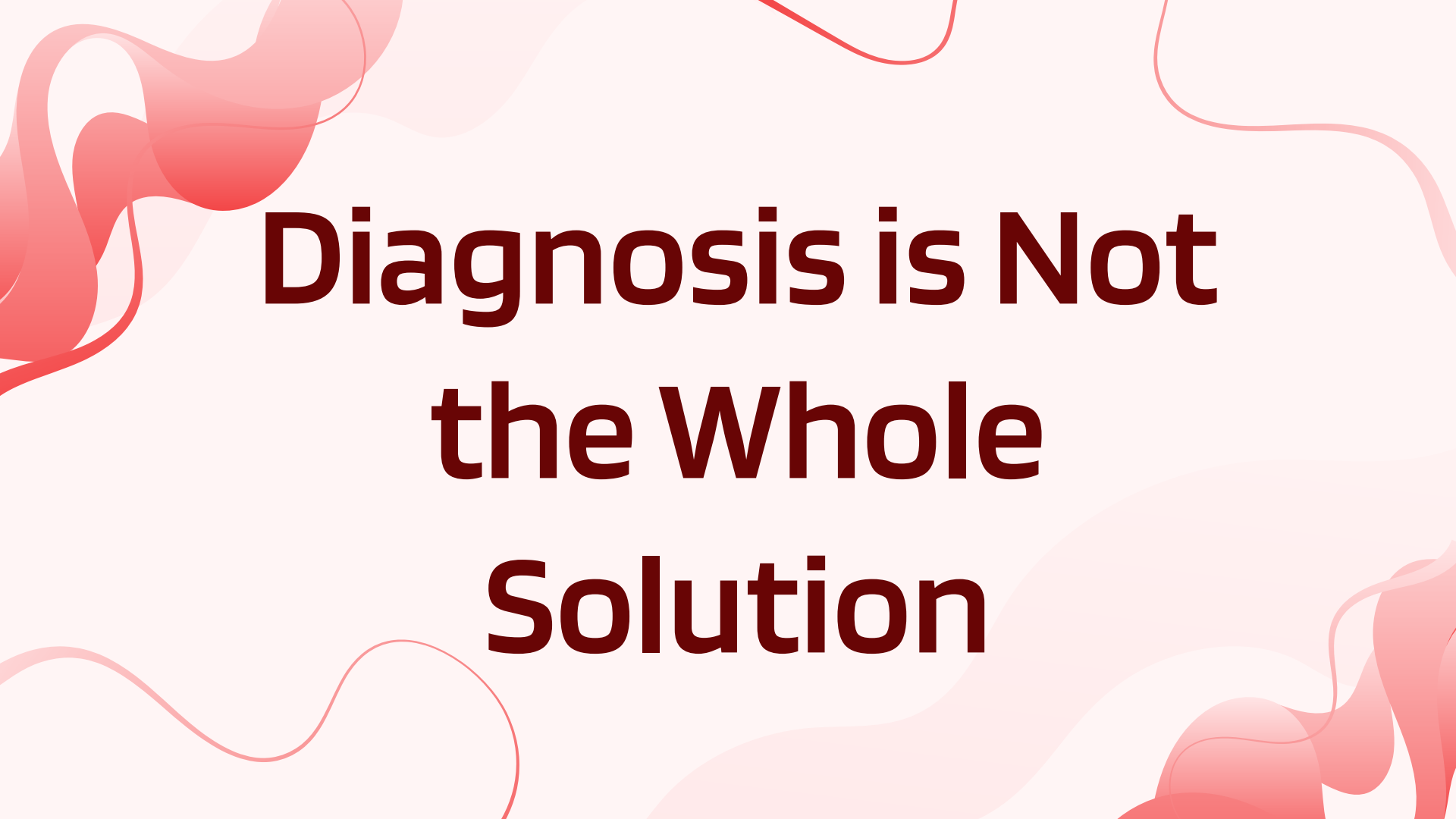
**50-90% of SCD Patients
die before the age of 5**
(*Grosse, 2011*)

**10% of all under 5 deaths in
Sub-Saharan Africa can be
attributed to SCD** (*Oron, et al., 2020*)

Immediate intervention needs to ensue in order to significantly improve child mortality and the quality-of-life for SCD patients.

Our Solution





**Diagnosis is Not
the Whole
Solution**

Develop research-based educational materials

Education, Advocacy, Policy

Symptoms
and
Screening

Referral
and
Diagnosis

Treatment
and
Management

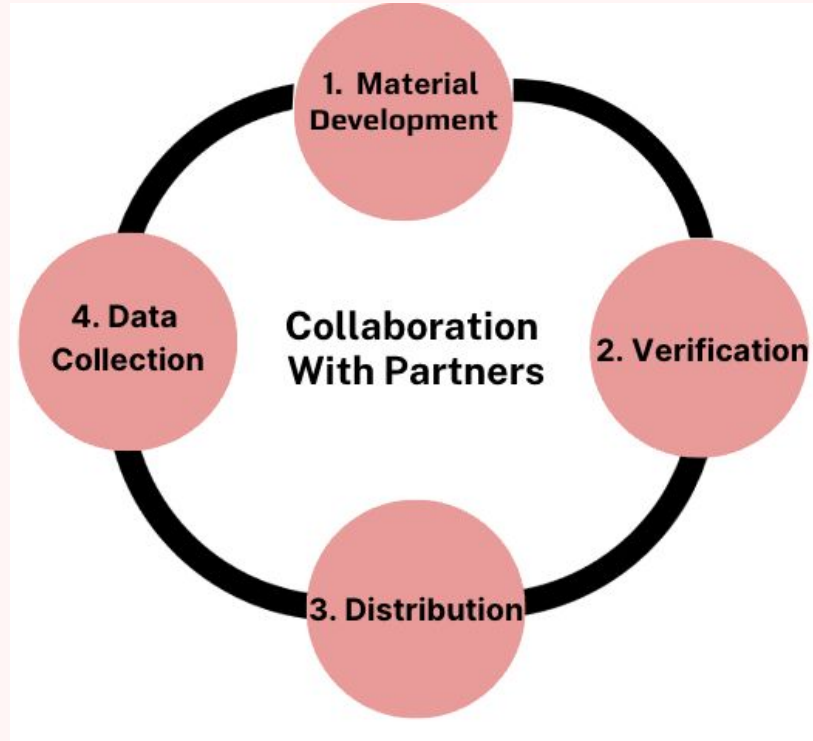
Monitoring

Deploy POC
test device

Develop
national
SOP

Data
collection
policy

Education and Advocacy



Capacity Building for Doctors and Nurses

1

With a **task force** of experts, develop outline that is accurate/relevant to context of Sierra Leone

REFINE

2

Partner with hospital to conduct **pilot course**, conduct **focus groups** to gather feedback and gauge efficacy

PILOT

3

Receive CME accreditation through CHASL and the **Ministry of Health**

ACCREDIT

4

Via healthcare system with assistance of NGOs, utilizing **workshops** and **training session** activities

IMPLEMENT

Module 1 Assessment

Quiz

1 / 5

What are some symptoms of sickle cell disease discussed in the text?

Fever and cough

Bone pain and anemia

Headache and dizziness

Complete and Continue >

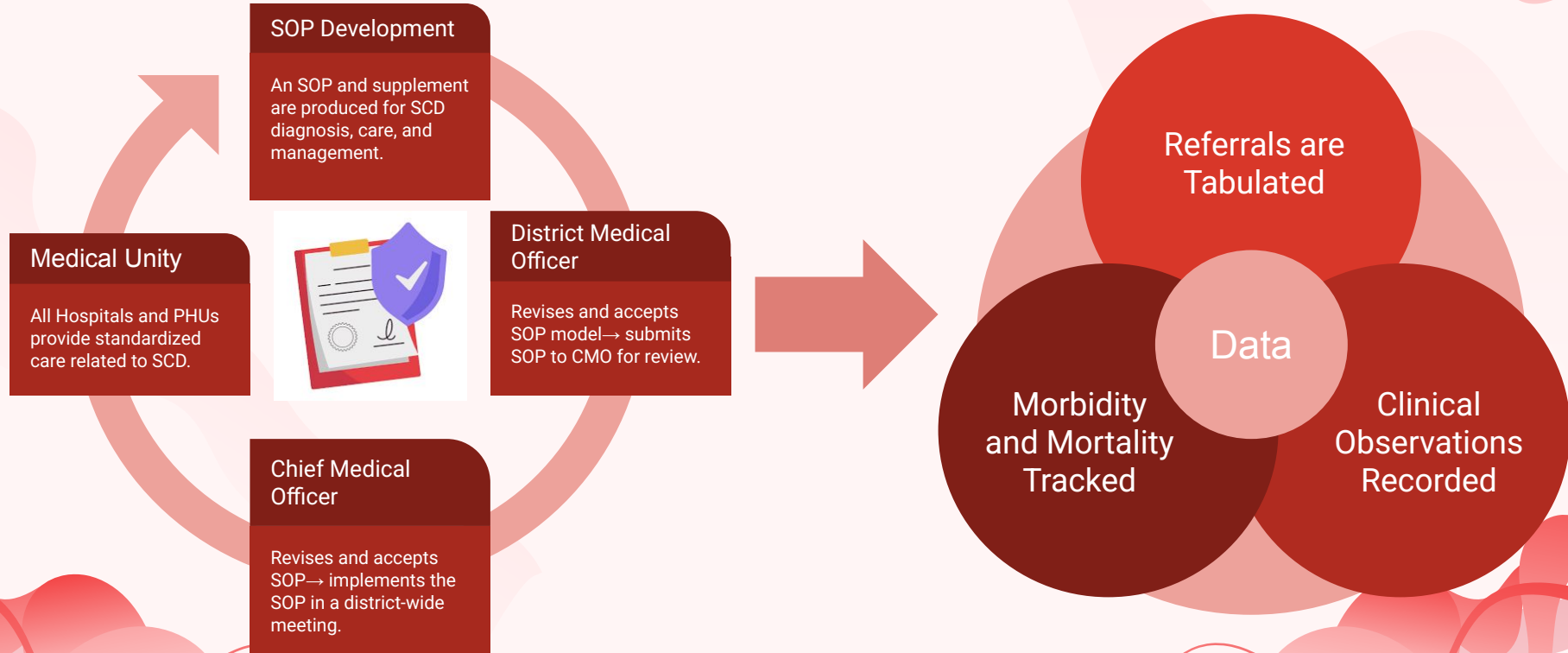
1.1 - PHUs and Referral

In rural communities in Sierra Leone, individuals often seek care at Primary Health Units (PHUs) for symptoms related to sickle cell disease such as bone pain and anemia. However, Community Health Workers (CHWs) in these areas may not always recognize these symptoms as being indicative of sickle cell disease. This lack of recognition can lead to a failure to refer patients to higher-level healthcare facilities, resulting in a delay in diagnosis and appropriate treatment for individuals with sickle cell disease.

It is crucial for CHWs to have a comprehensive understanding of the signs and symptoms of sickle cell disease. By being able to recognize these manifestations in patients, CHWs can promptly refer individuals to hospitals for further evaluation and more specialized care. Training CHWs to identify sickle cell disease early can significantly impact patient outcomes, as timely interventions can help in managing the symptoms and complications associated with the disease.

Empowering CHWs with the knowledge and skills to identify and refer individuals with sickle cell disease is essential in improving the overall management of the condition in Sierra Leone. Through targeted educational programs and continuous support, CHWs can

Standardization of Care



CUSTOMERS AND CONSUMERS

SL Ministry of
Health / Pharmacy
Board



Healthcare
facilities



Healthcare
providers



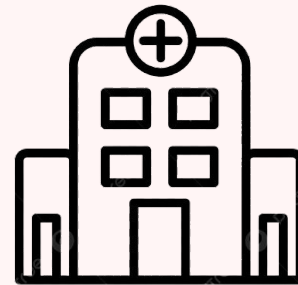
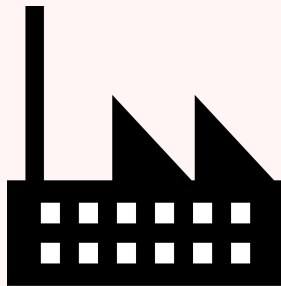
Patients

REVENUE STREAMS

Initial funding



Steady state funding



Diagnosis free of charge for patients



Metrics for Success

Short



Increased
understanding
of SCD



Establish
trust &
relationships

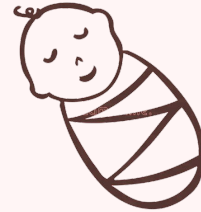


Test
Usage/newborn
screening

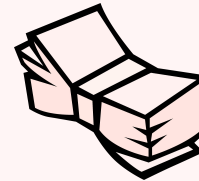


Generalization
across other
LMICs

Long



Decrease
under-5 child
mortality rate



Cost
savings

SickLED 2024 Accomplishments

- Device Optimization
- Identifying and Reaching out to Contacts
- Fieldwork Plan Brainstorming

**Spring 2024
Fieldwork
preparation**

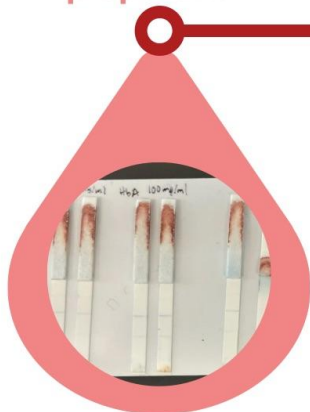


- Four Point Plan Development
- Formation of Key Partnerships
- Resource Refining

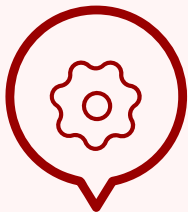
Fieldwork 2024

Mountaintop 2024

- Device Optimization
- IRB Document Completion
- Flipbook Finalization



Future Plans



2024-2025 OPTIMIZE

Optimize our device and begin searching for an ethical manufacturer.



2024-2026 EDUCATE

Work with local organizations to boost the local understanding of SCD. Educate healthcare providers on our device.



2025-2026 IMPLEMENT

Implement our device with support of outside funding and the Pharmacy Board. Work to cover the test under the Free Healthcare Initiative.



2026 MAINTAIN

Hand-off the project to in-country partners to sustain its development. Work on expanding to other countries.

Thank you!

SICKLED

Advocacy for Sickle Cell Education and Diagnosis

Stay connected with us:



FAQs

1. What do our educational resources look like?
2. Who are our competitors?
3. Science of the test
4. What complications does sickle cell cause?
5. What are the current diagnosis options for sickle cell disease?
6. Sickle cell disease in Sierra Leone vs the US
7. More about global partners

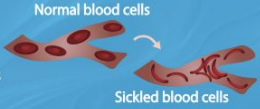
POSTERS

SICKLED
Initiated by Sickle Cell Movement and Lehigh

**LEHIGH**
CREATIVE INQUIRY

What is Sickle Cell Disease?

-A genetic condition where red blood cells take on a sickled shape, blocking blood flow and causing bone pain and other complications
-It is inherited from both parents



Normal blood cells
Sickled blood cells

Common Symptoms



Tiredness



Stunted growth



Yellow eyes and skin



Bone pain



Dizziness

HOW TO MANAGE SICKLE CELL DISEASE

What To Do ✓



Go to the nearest facility for medical advice



Drink plenty of water



Balanced diet



Stay warm

What To Avoid ✗



Avoid alcohol and smoking

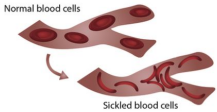


Avoid overworking



Avoid tying yourself up

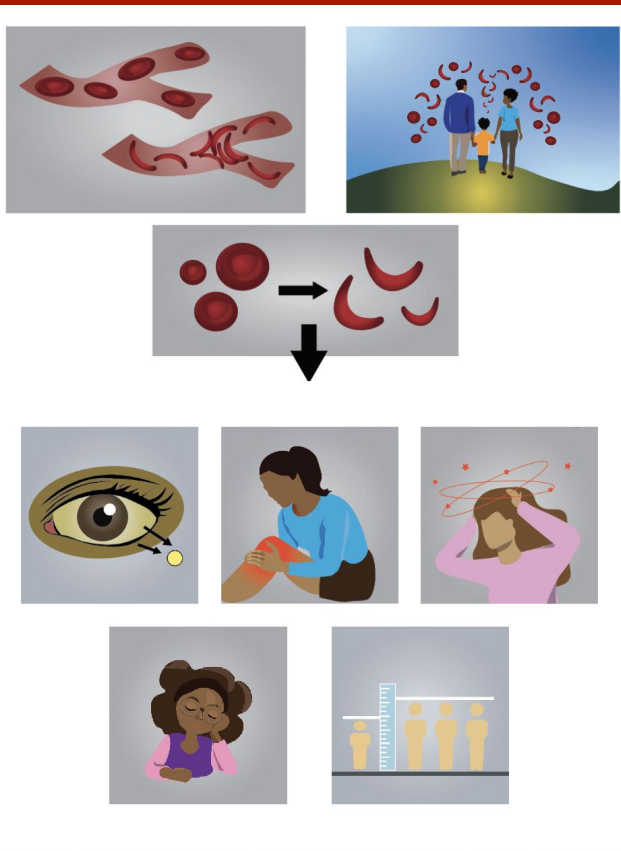
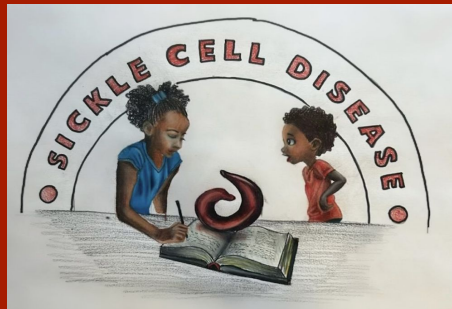
School Resources

FRONT PAGE	BACK OF PAMPHLET	FIRST FLAP
 LEHIGH UNIVERSITY  A Teacher's Guide to Sickle Cell Disease What is Sickle Cell Disease? Sickle Cell Disease is a genetic condition inherited from both parents where red blood cells take on a sickled shape. This blocks blood flow, causing bone pain and other complications. Different forms of the disease vary in severity; some cause more complications than others. 		Common Signs and Symptoms: Shortness of breath Tiredness Yellow eyes: An increased breakdown of red blood cells produce bile, turning the eyes yellow. Pale skin/nails/tongue: Destruction of red blood cells cause anemia. Pain in bones/joints: Sickled shape of cells Results in a block of blood flow causing Pain. Sores on ankles: Sores develop from poor blood flow, infection, or swelling. • NONE of these symptoms are contagious • If a child is experiencing a crisis, or their condition changes, bring them to a doctor for help.

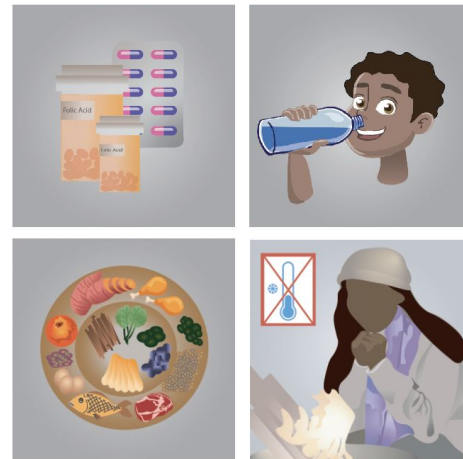
Left Interior	Middle Interior	Right Interior
A Teacher's Role in Sickle Cell Disease: Sickle cell disease can affect how the student functions in school so teachers need to be aware of the signs and symptoms, misconceptions, and how to handle a sickle cell student. Promoting an Inclusive Environment 1. Discuss Sickle Cell disease in class to increase general awareness 2. Create a sickle cell club providing an outlet for children across ages 3. Allow discretion for students to approach and discuss sickle cell disease in a private setting 4. Do not scold children with sickle cell for resting or following management protocols	Considerations in School: If a child misses school... Children with sickle cell disease frequently miss school due to experiencing a crisis or going to the doctor. Do not punish them for these absences. If a child is teased... Children can be very unkind to children who are different. Children with sickle cell can feel ashamed from teasing and not want to attend school as a result. Teaching students about sickle cell can encourage students to be understanding and kinder to one another. If a child asks to go to the bathroom... Children with sickle cell disease are encouraged to drink lots of water as it is important for the management of the disease. Because of this, children may use the bathroom frequently. Do not discourage this. If a child is tired... Anemia can make a child tired. Let them put their head down on the table or lay down and rest.	How to Manage a Child: Recognize Symptoms: Be alert to signs like complaints of bone pain, yellowing eyes, continuous sickness, or fatigue. Give Plenty of Water: Ensure the child has plenty of water throughout the day. Respond to Changes: Seek medical advice if a child reports feeling sick every day. If a child's condition worsens, immediately send them to the hospital. Physical education: Allow children to take part in physical activities. If they experience pain or shortness of breath, allow them to rest. Common Misconceptions: It IS possible to live into adulthood with proper management BOTH parents contribute to the expression of the disease. It is NOT witchcraft It is NOT cured by herbal medicines It is NOT healthy to constrict the patient's limbs



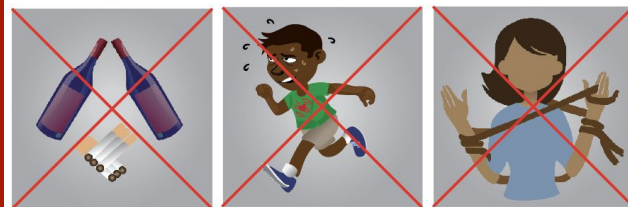
Flipbook



Do ✓



Don't ✗



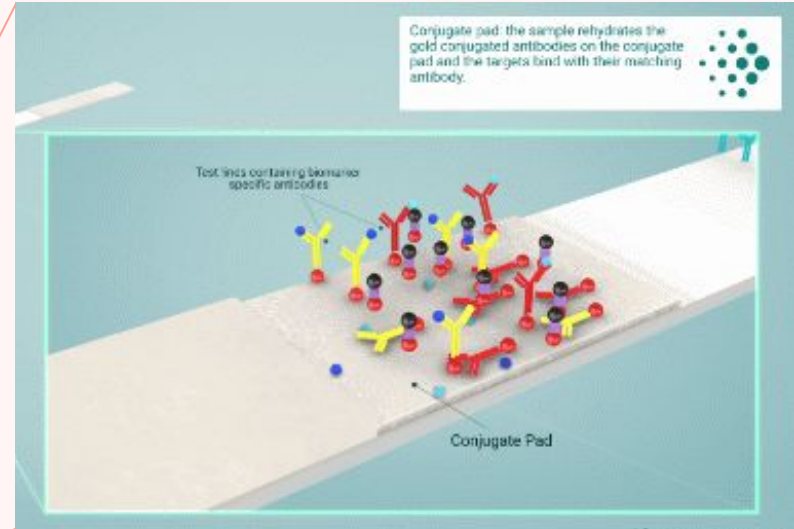
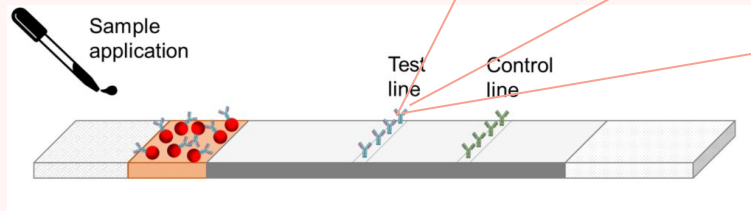
Competitors

Sickle Scan:

- Discontinued in Sierra Leone
- Expensive (\$7 per test)
 - Dilution step
- Plastic packaging
- Not sustainable



Sandwich Assay

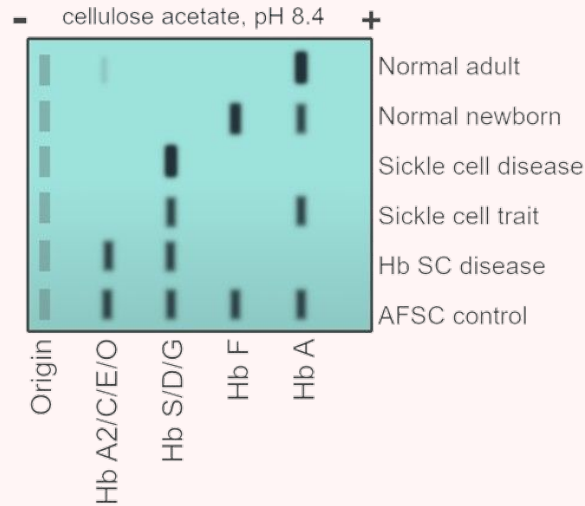


Diagnosis Options

Sickle Solubility Test



Hemoglobin electrophoresis



Hemoglobin electrophoresis

HPLC



Sickle Cell Disease Complications

Streptococcus
Pneumoniae

Acute Chest Syndrome

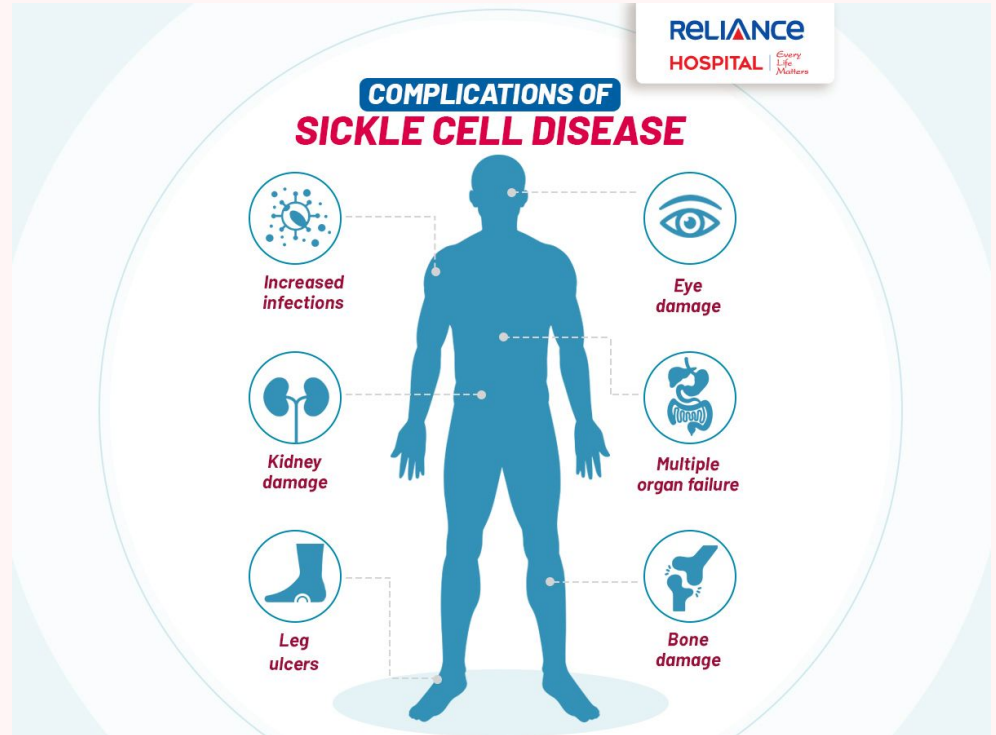
Bone Damage

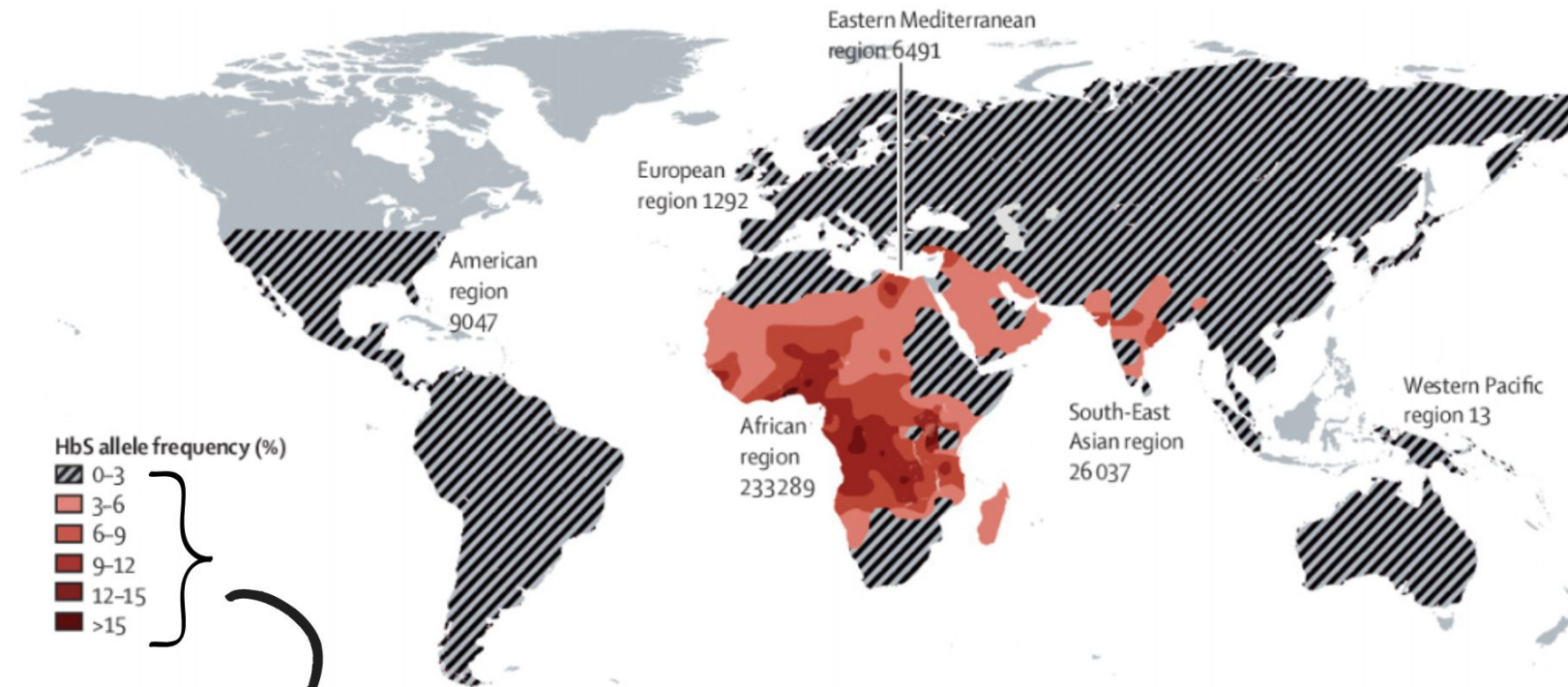
Organ damage

Blood Clots

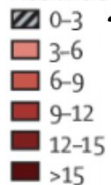
Stroke

Kidney Damage





HbS allele frequency (%)

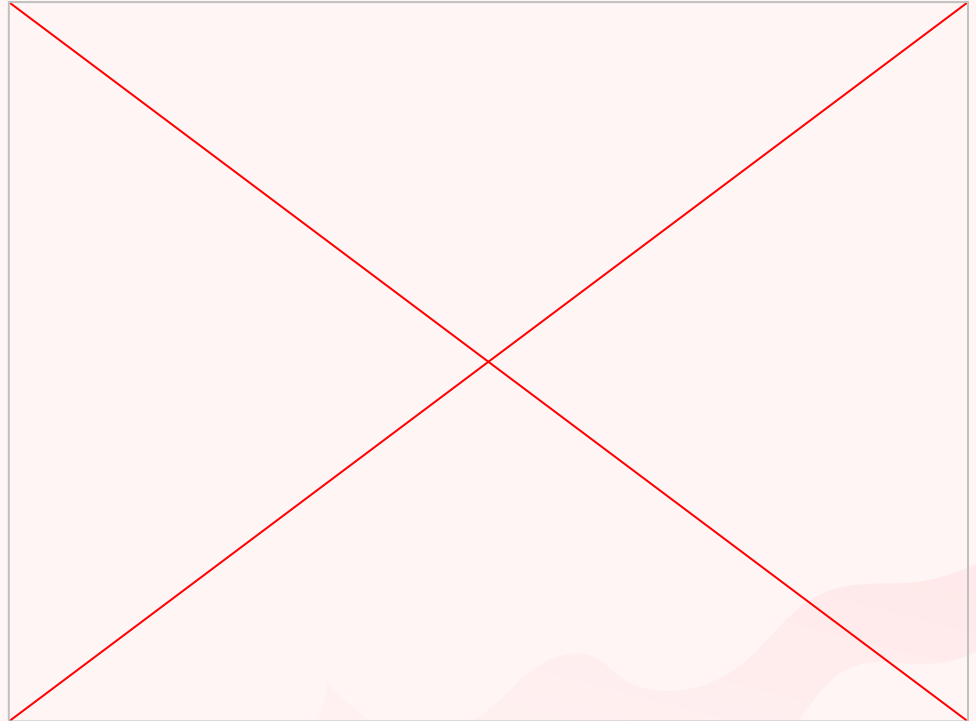


Estimates # of affected individuals per year.

Global Partners

Mr. Abdulai Ibrahim Sankoh

Founder of Sankoh Humanitarian
Advocacy for Sickle Cell (SHAFS)



Revenue Models

Budget of non-recurring expenses

Expenses	Period 1	Period 2	Period 3	Period 4	Period 5	Period 6
Networking Expenses	\$ 7,500.00	\$ 7,500.00	\$ 6,000.00	\$ 6,000.00	\$ 5,000.00	\$ 5,000.00
Communications (Top-up, Internet)	\$ 2,000.00	\$ 2,000.00	\$ 2,000.00	\$ 2,000.00	\$ 2,000.00	\$ 2,000.00
Transportation Costs	\$ 3,000.00	\$ 3,000.00	\$ 3,000.00	\$ 3,000.00	\$ 3,000.00	\$ 3,000.00
Advertising Expenses	\$ 15,000.00	\$ 10,000.00	\$ 5,000.00	\$ 5,000.00	\$ -	\$ -
Misc. Startup expenses	\$ 80,000.00	\$ 70,000.00	\$ 25,000.00	\$ 20,000.00	\$ 15,000.00	\$ 5,000.00
Total non-recurring expenses	\$ 107,500.00	\$ 92,500.00	\$ 41,000.00	\$ 36,000.00	\$ 25,000.00	\$ 15,000.00

Typical cost breakdown for lateral flow devices (per test)

Material	Cost Estimate on Manufacturing Scale
Test Line antibodies	\$0.01 x 2
Membranes	\$0.02
Control Antibodies	\$0.01
Absorbance Pad	\$0.001
Conjugate Antibodies	\$0.02
Sample Pad	\$0.002
Latex Microspheres	\$0.01
Membrane Backing	\$0.01
Desiccant	\$0.02
Pouch	\$0.08
Plastic Housing	\$0.20

Labor estimated at twice the cost of raw materials

Total \$0.39 (likely too high)

Part Number

Part Name Description

Unit Cost

1	FF80HP Nitroce	Where the antibodies will be deposited and where the beads	\$0.11
2	CF5 Absorption	Where the sample and beads will be collected at the end of	\$0.20
3	Standard 17 Gl	Where the blood sample will be placed on the strip, wash st	\$0.23
4	400 nm Blue LA	Beads that will be conjugated with anti-HbA antibody to det	\$1.13
5	Rockland: Anti-I	HbA Test Line Antibody	\$2.90
6	Donkey anti-Go	Control Line Antibody	\$0.35
7	Polyclonal goat	Antibody on beads	\$0.01
9	Anti-HbS	HbS Test Line (Approximation based on HbA Test Line)	\$2.40
10	Slide-A-Lyzer™	Used to dialyze rabbit polyclonal anti-HbA antibody, remove	\$0.20
11	Lamination		\$0.01

TOTAL COST PER STRIP

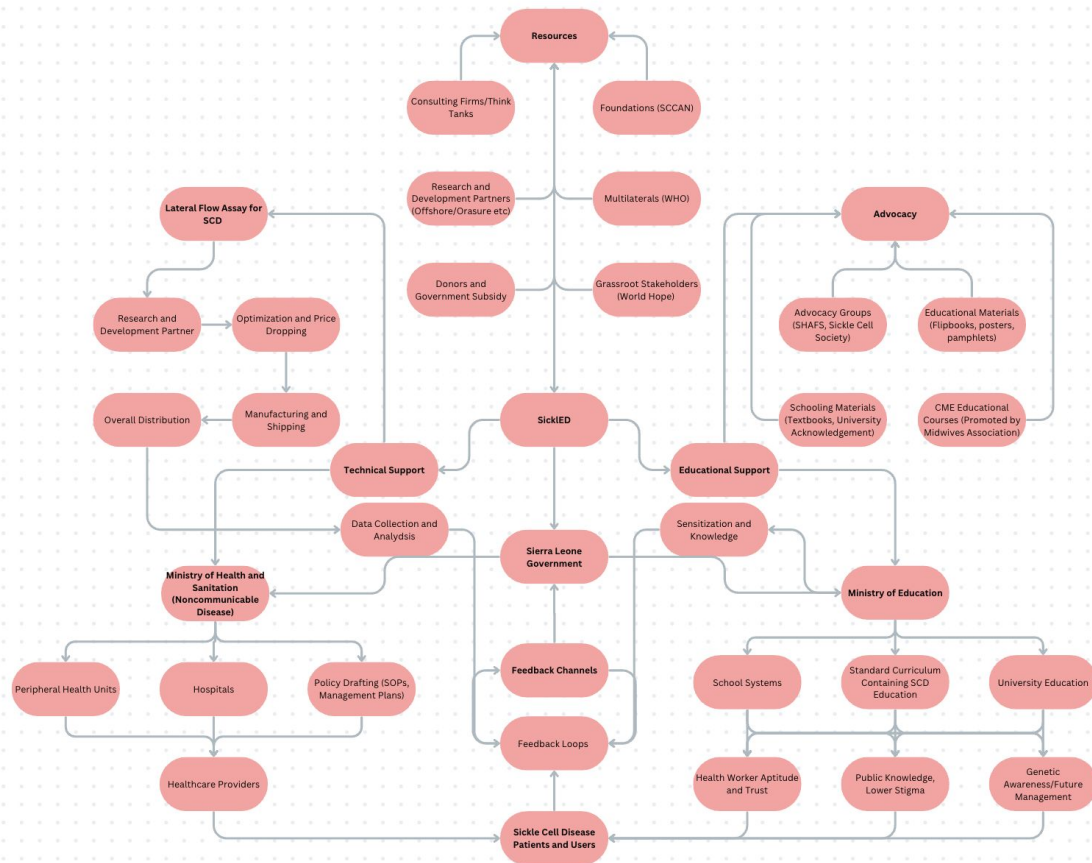
\$ 7.540 For the current testing models in lab without scale.

Revenue Models

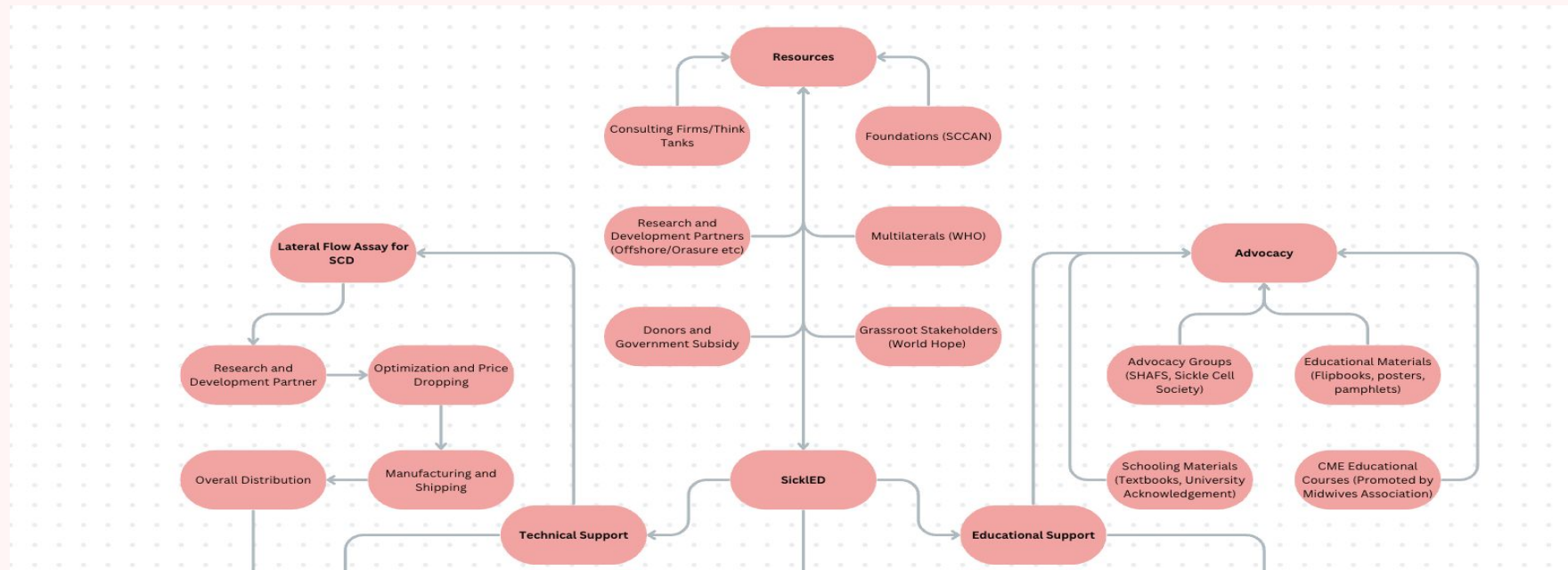
		SickLED					
		Pro Forma Income Statement for Periods 1-6					
		Period 1	Period 2	Period 3	Period 4	Period 5	Period 6
Variable*** Variable***	Total Revenues	\$ 115,250.00	\$ 165,750.00	\$ 286,500.00	\$ 393,000.00	\$ 567,000.00	\$ 673,500.00
	Total Unit Sales	25,500	36,500	63,000	86,000	124,000	147,000
	COGS:						
	Raw Materials	\$ 12,750.00	\$ 18,250.00	\$ 31,500.00	\$ 43,000.00	\$ 62,000.00	\$ 73,500.00
	Packaging Costs	\$ 5,100.00	\$ 7,300.00	\$ 12,600.00	\$ 17,200.00	\$ 24,800.00	\$ 29,400.00
	Shipping Costs	\$ 6,375.00	\$ 9,125.00	\$ 15,750.00	\$ 21,500.00	\$ 31,000.00	\$ 36,750.00
	Total COGS	\$ 24,225.00	\$ 34,675.00	\$ 59,850.00	\$ 81,700.00	\$ 117,800.00	\$ 139,650.00
	Gross Profit	\$ 91,025.00	\$ 131,075.00	\$ 226,650.00	\$ 311,300.00	\$ 449,200.00	\$ 533,850.00
	Staffing Expenses						
	Project Manager Salary	\$ 20,000.00	\$ 20,000.00	\$ 20,000.00	\$ 20,000.00	\$ 22,000.00	\$ 22,000.00
Rough Estimate	Project Manager Bonuses	\$ 5,000.00	\$ 5,000.00	\$ 6,000.00	\$ 6,000.00	\$ 7,500.00	\$ 7,500.00
	Relationship Manager Salary	\$ 9,000.00	\$ 9,000.00	\$ 9,000.00	\$ 9,000.00	\$ 10,000.00	\$ 10,000.00
	Training Expert	\$ 6,000.00	\$ 6,000.00	\$ 6,000.00	\$ 6,000.00	\$ 6,000.00	\$ 6,000.00
	Other Expenses						
	Networking Expenses	\$ 7,500.00	\$ 7,500.00	\$ 6,000.00	\$ 6,000.00	\$ 5,000.00	\$ 5,000.00
	Communications (Top-up, Internet)	\$ 2,000.00	\$ 2,000.00	\$ 2,000.00	\$ 2,000.00	\$ 2,000.00	\$ 2,000.00
	Transportation Costs	\$ 3,000.00	\$ 3,000.00	\$ 3,000.00	\$ 3,000.00	\$ 3,000.00	\$ 3,000.00
	Advertising Expenses	\$ 15,000.00	\$ 10,000.00	\$ 5,000.00	\$ 5,000.00	\$ -	\$ -
	Misc. Startup Expenses	\$ 80,000.00	\$ 70,000.00	\$ 25,000.00	\$ 20,000.00	\$ 15,000.00	\$ 5,000.00
	Total Operating Expenses	\$ 147,500.00	\$ 132,500.00	\$ 82,000.00	\$ 77,000.00	\$ 70,500.00	\$ 60,500.00
Break-Even Point (Units)		41,321	36,897	22,793	21,272	19,461	16,659
Net Income		\$ (56,475.00)	\$ (1,425.00)	\$ 144,650.00	\$ 234,300.00	\$ 378,700.00	\$ 473,350.00

Selling Price per Unit (Direct Channels)	\$ 5.50
Bulk Pricing (Indirect Channels)	\$ 4.50
COGS:	
OEM Sickle Cell Test	\$ 0.50
Shipping/Transportation	\$ 0.25
Packaging Costs	\$ 0.20

Animated Visual of Redesigned System



Animated Visual of Redesigned System



Animated Visual of Redesigned System

